

Harpreet Singh

Computer Engineering Student
AI Engineering | Machine Learning | Backend Systems

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EDUCATION

University of British Columbia (UBC)
BASc, Computer Engineering (Vancouver, BC)

Sept 2024 to Dec 2027 (expected)

EXPERIENCE

Software Development Engineer Intern, Amazon

May 2026 to Jul 2026

- Developing production software in a large-scale engineering environment, contributing to implementation, debugging, testing, and code review.
- Applying backend and systems fundamentals to build maintainable code and integrate with existing software architecture.

MOps Software Developer, UBC ORBIT Satellite Design Team

Jan 2026 to Present

- Building mission-operations software across frontend, backend, deployment, and testing workstreams.
- Working with other ORBIT subteams to translate engineering requirements into tools, API integrations, and reliable internal workflows.

PROJECTS

RAG Retrieval Pipeline for Computer Systems QA

Academic Team Project

Python | PyMuPDF | all-MiniLM-L6-v2 | FAISS | JSONL | Unit Tests

- Built the retrieval layer for a RAG-style QA system: preprocessing, page cleaning, chunking, embedding, vector indexing, and top-k search.
- Chunked 8 source documents into 11,890 MiniLM-ready chunks and built a 384-d normalized FAISS index for semantic retrieval.
- Implemented query-time retrieval that embeds questions, searches FAISS, and returns ranked source/page metadata for grounded answer generation.
- Wrote unit tests for chunk overlap, source separation, and long-page splitting to keep retrieval metadata reliable.

AI Agents and MCP Exploration

Self-Directed Project

Google ADK | MCP | A2A | Tool Calling | Agent Workflows

- Built hands-on agent prototypes while studying Google Agent Development Kit concepts, MCP tool integration, and Agent-to-Agent communication patterns.
- Followed Hugging Face MCP course material to understand tool servers, external data/tool access, structured calls, and agent workflow design.

CNN Number Classification

Academic Project

Python | PyTorch | CNNs | Computer Vision | Distribution Shift

- Trained a CNN on 28x84 grayscale number images using Conv2d, batch normalization, ReLU, max pooling, dropout, Adam, and cross-entropy loss.
- Built training/evaluation loops with model checkpointing, test accuracy reporting, and distribution-shift evaluation across multiple shifted datasets.

DNN Reacher Policy from Demonstrations

Academic Project

Python | PyTorch | Imitation Learning | AdamW | SmoothL1Loss

- Trained a neural policy on reacher demonstration data using normalized 6-dimensional observations and 2-dimensional continuous actions.
- Implemented a 4-layer DNN with LayerNorm, ReLU, Tanh output, validation loss tracking, learning-rate scheduling, gradient clipping, and checkpointing.

Player Engagement Analysis

Academic

R | Statistical Modeling | KNN | Regression

- Analyzed engagement patterns with KNN and regression models to support capacity/resource planning decisions.

SKILLS

Languages: Python, Java, TypeScript, R, SQL, C++
Backend: Spring Boot, REST APIs, PostgreSQL, Node.js

ML/AI: PyTorch, CNNs, Embeddings, FAISS, MCP, Google ADK
Tools: Git/GitHub, Docker, NumPy, Matplotlib, VS Code